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PROJECT TITLE

AR-based & AI-Powered Smart Heritage Tourism Platform

ABSTRACT

This project addresses the limitations of traditional tourism methods in Penang, where visitors rely on static guidebooks and signage that provide limited interactivity and personalization. As a result, tourists may not fully engage with or understand the historical value of heritage sites. To overcome this issue, an AR-based and AI-powered smart heritage tourism platform is proposed to enhance tourist engagement and experience. The system integrates Augmented Reality (AR) to display interactive 3D models and historical content, while Artificial Intelligence (AI) provides personalized recommendations, visual recognition, and real-time itinerary optimization. The platform consists of a mobile application for tourists and a web-based system for service partners and administrators. Additional features include smart navigation, booking and payment integration, and analytics dashboards. The expected outcome is a more immersive and personalized tourism experience, improved accessibility to cultural information, and enhanced data-driven tourism management. (141 words)

PROBLEM

Traditional tourism in heritage-rich locations such as Penang still relies heavily on static information sources such as printed guidebooks, signage, and basic mobile applications. These methods provide limited interactivity, lack real-time updates, and do not offer personalized experiences for tourists. As a result, visitors often miss important historical context and may not fully engage with cultural heritage sites. This limitation is particularly significant in heritage tourism, where understanding historical context plays a crucial role in enhancing visitor experience. Studies have shown that Augmented Reality (AR) can significantly enhance user engagement and improve the tourism experience by providing interactive and immersive content (tom Dieck & Jung, 2015; Cranmer et al., 2020).

Furthermore, tourists often face difficulties in planning efficient itineraries due to the lack of intelligent systems that consider user preferences and real-time conditions. Most existing tourism platforms provide general recommendations rather than personalized suggestions, leading to less effective travel planning. Artificial Intelligence (AI) has been widely recognized as a key technology that enables personalized recommendations and improves tourism experiences through data-driven decision-making (Ozcelik et al., 2023).

In addition, tourism stakeholders and heritage site administrators have limited access to real-time data and analytics, making it difficult to manage visitor flow and improve tourism services. The use of digital technologies in tourism, including AR and AI, has been shown to enhance operational efficiency and support better tourism management (Obeidy et al., 2018; Fong & Lau, 2025). Furthermore, immersive and virtual technologies have been identified as promising solutions for cultural heritage promotion in the Southeast Asian context (Michael, 2025; Garbin

Praničević, 2021), and AI-driven systems have demonstrated measurable improvements in both guide experiences and visitor satisfaction (Harris, 2025; Sánchez-Martín et al., 2025).

Therefore, with the advancement of immersive technologies and intelligent systems, there is a need for a smart tourism solution that enhances user engagement, provides personalized experiences, and supports data-driven tourism management.

SOLUTION

This project proposes the development of an AR-based and AI-powered smart heritage tourism platform to address the limitations of traditional tourism methods in Penang. The system consists of a cross-platform mobile application for tourists and a web-based application for service partners and administrators. This design ensures that each group of users can access relevant features efficiently while supporting a complete tourism ecosystem. The system adopts a dual-platform architecture consisting of a mobile application for end users and a web-based management system for service partners and administrators.

The mobile application enhances tourist experience through the use of Augmented Reality (AR), which allows users to view interactive 3D models, historical overlays, and multimedia content at heritage sites. This improves tourists' understanding of cultural and historical information and increases engagement.

In addition, Artificial Intelligence (AI) is integrated into the system to provide personalized recommendations, perform visual recognition of heritage sites, and generate optimized travel itineraries. By considering user preferences and real-time factors such as weather and crowd density, the system helps tourists plan their trips more efficiently.

The platform also includes smart navigation features with real-time route guidance and context-aware suggestions, addressing inefficient itinerary planning. Furthermore, integrated booking and payment services allow users to manage their travel activities in one system, while user-friendly features enhance overall engagement. Analytics dashboards are provided for service partners and system administrators to monitor visitor trends and support data-driven decision-making.

Critical Functions:

- User registration, authentication, and profile management
- AR-based heritage exploration with interactive 3D models and multimedia content
- AI-powered personalized recommendations and visual recognition
- AI-based itinerary generation and travel planning
- Smart navigation with real-time guidance and interactive maps
- Booking and reservation management for tourism services
- Payment processing and transaction management
- Review and feedback system for users
- Safety, accessibility, and notification support
- Service partner account and service management
- Pricing and booking management for service partners
- Financial reporting and basic analytics for service partners
- Administration for managing users and service partners
- Booking, transaction, and commission management

- Analytics dashboard for monitoring and decision-making

TARGET MARKET

The target users of this system include tourists, service partners, and system administrators, each interacting with the platform according to their roles.

Tourists, including both local and international visitors, are the primary users. They will use the mobile application to explore heritage sites through Augmented Reality (AR), access interactive 3D models and historical content, receive personalized recommendations, generate travel itineraries, and navigate efficiently using smart maps. They can also make bookings, complete payments, interact with platform features and share their experiences.

Tourism service partners, such as tour operators, accommodation providers, restaurants, and transportation services, will use the web-based system to manage their services, update pricing and availability, handle bookings, respond to feedback, and monitor performance through analytics dashboards.

System administrators will manage the overall platform, including user and partner management, AR content maintenance, system monitoring, and report generation to support decision-making.

Overall, the system connects tourists, service partners, and system administrators into a unified tourism ecosystem.

COMPETITION / CONTRIBUTION

Existing platforms such as Google Maps and TripAdvisor mainly provide navigation, reviews, and general travel information, but they rely on static content and offer limited interactivity and personalization. More specialized platforms such as izi.Travel offer audio-guided heritage tours, while trip planning platforms such as Trip.com provide itinerary management, yet none of these platforms integrate AR-based site visualization with AI-driven personalization in a single unified system. Local heritage applications in Penang similarly offer basic information browsing without immersive or intelligent features.

The proposed system is unique as it integrates AR and AI into a single platform. AR enables interactive 3D visualization of heritage sites, while AI provides personalized recommendations and real-time itinerary optimization. Features such as AR navigation and integrated booking services further enhance the user experience.

The system offers advantages including improved engagement, personalized travel planning, and efficient tourism management. It supports data-driven decision-making through analytics dashboards.

In terms of contribution, the system enhances tourist experience, supports service partners in improving business performance, and promotes cultural heritage awareness. It also contributes to local economic growth by driving heritage-based tourism.

MILESTONES

Proposed Development Model / Research Method :

This project adopts the Evolutionary Prototyping model, which begins with the development of an initial working prototype that is continuously refined and extended based on feedback and testing until it evolves into the final system. This model is particularly suitable for this project due to the experimental nature of Augmented Reality (AR) and Artificial Intelligence (AI) components, where requirements cannot be fully defined upfront. Rather than discarding early versions, each iteration builds upon the previous one, progressively incorporating more features and improvements. The model involves four key phases repeated across iterations: prototype development, evaluation, refinement, and expansion. This ensures that complex integrations such as AR heritage exploration, AI-powered recommendations, smart navigation, and booking systems are validated and improved continuously throughout the development lifecycle, resulting in a robust and user-centered final system.

Project Schedule:

ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Proposal Approval & Title Finalization	Approved project title and proposal	13/03/2026
Proposal Submission (Form 1 - 3)	Completed and submitted proposal	03/04/2026
Chapter 1 : Introduction	Completed introduction chapter	26/06/2026
Chapter 2: Research Background	Completed literature review	10/06/2026
Chapter 3: Methodology & Requirements	Completed system methodology, requirements analysis and initial prototype design	24/07/2026
Chapter 4: System Design	Completed system architecture, UI design and Prototype 1(core AR and user module)	14/08/2026
FYP1 Portfolio Submission	Completed report (Chapter 1 - 4) with Prototype 1	04/09/2026
Initial Prototype Evaluation	Feedback collected for prototype improvement	14/09/2026
Prototype 2 Development & Evaluation	Refined prototype with AI recommendation, navigation and booking modules integrated	16/09/2026
System Preview II	Evaluation of improved prototype with test plan	02/11/2026
Prototype 3 Refinement & Expansion	Enhanced prototype with payment, partner and admin modules	11/11/2026
Final System Testing & Validation	Completed system testing, bug fixed and user feedback incorporated	18/11/2026
Draft Report Submission	Submitted draft FYP report with complete system	04/12/2026
Final Report & Deliverables	Final evolved system, complete report and source code submitted	11/12/2026

REFERENCES

1. tom Dieck, M. C., & Jung, T. (2015). A theoretical model of mobile augmented reality acceptance in urban heritage tourism. *Current Issues in Tourism*, 21(2), 154–174. <https://doi.org/10.1080/13683500.2015.1070801>
2. Cranmer, E. E., tom Dieck, M. C., & Fountoulaki, P. (2020). Exploring the value of augmented reality for tourism. *Tourism Management Perspectives*, 35, 100672. <https://doi.org/10.1016/j.tmp.2020.100672>
3. José-Manuel Sánchez-Martín, Guillén-Peñafiel, R., & Ana-María Hernández-Carretero. (2025). Artificial Intelligence in Heritage Tourism: Innovation, Accessibility, and Sustainability in the Digital Age. *Heritage*, 8(10), 428–428. <https://doi.org/10.3390/heritage8100428>
4. Garbin Praničević, D. (2021). Augmented Reality and Virtual Reality-Based Technology in Cultural Tourism. *ENTRENOVA - ENTERprise REsearch INNOVation*, 7(1), 314–322. <https://doi.org/10.54820/mhny8236>
5. Suayb Talha Ozcelik, Meltem Turhan Yondem, Caetano, I., Figueiredo, J., Alves, P., Goreti Marreiros, Huseyin Bahtiyar, Yuksel, E., Perales, F., & Suci, G. (2023). *Transforming Tourism Experience: AI-Based Smart Travel Platform*. <https://doi.org/10.1145/3651640.3651645>
6. Michael, P. (2025, July 17). *Virtual and Immersive Technologies in Cultural Heritage Tourism | Swinburne University, Sarawak, Malaysia*. Swinburne University, Sarawak, Malaysia | Swinburne University of Technology | Sarawak | Malaysia. <https://www.swinburne.edu.my/swinsays/virtual-and-immersive-technologies-in-cultural-heritage-tourism/>
7. Harris, M. (2025, September 17). *Enhancing Tour Guide experiences with AI and AR - Globibo Blog*. Globibo Articles & Research. <https://globibo.blog/enhancing-tour-guide-experiences-with-ai-and-ar/>
8. Waqas Khalid Obeidy, Arshad, H., & Huang, J. Y. (2018). TouristicAR: A Smart Glass Augmented Reality Application for UNESCO World Heritage Sites in Malaysia. *Journal of Telecommunication, Electronic and Computer Engineering (JTEC)*, 10(3-2), 101–108. <https://jtec.utem.edu.my/jtec/article/view/4719>
9. Fong, C. K., & Lau, K. S. (2025). Bridging Technology and Heritage: Mobile AR Applications as Efficient Tools for Cultural Destination Promotion. *Asia-Pacific Journal of Innovation in Hospitality and Tourism*, 14(1), 1–21. https://fslmjournals.taylors.edu.my/wp-content/uploads/APIHT-141_Paper-1.pdf